

BOARD BRIEFING · MAREC INSIGHTS

Convention Revenue Leakage Intelligence **System**

How MAREC Insights re-engineered margin visibility across the convention-services portfolio — from a –11.51% loss position to a validated \$1.01M recovery plan.

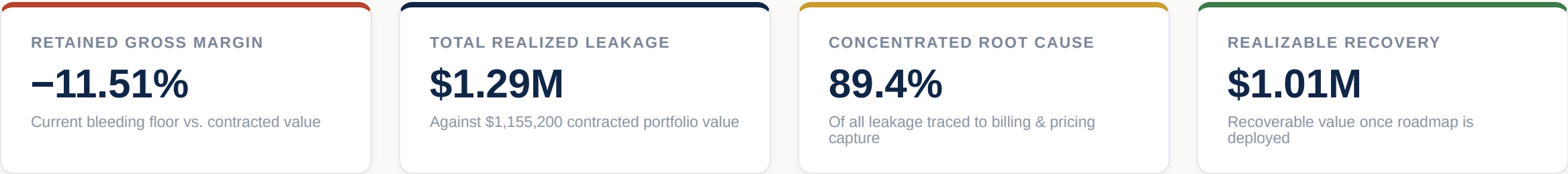
BUILT IN PYTHON

MODELED IN POWER BI

The Baseline > The Framework > The Diagnosis > The Roadmap > The Recovery

The Portfolio at a Glance

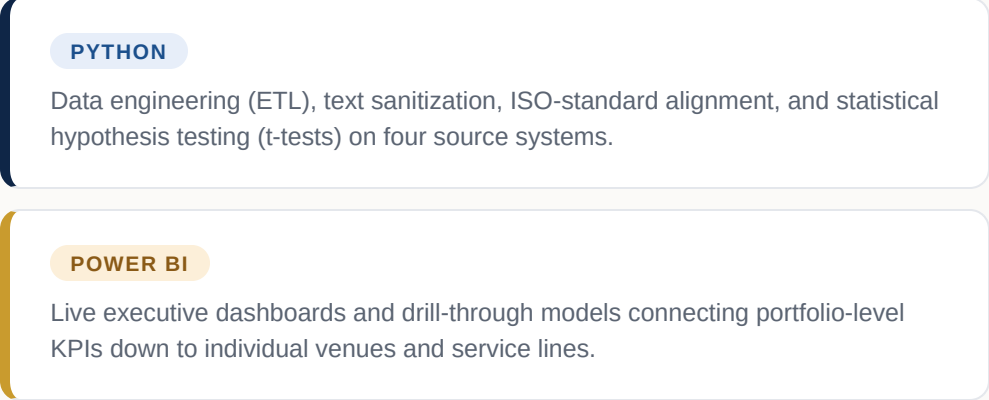
A single governance engine now traces every dollar of margin loss from raw operational logs to a prioritized, board-ready recovery plan.



What the board needs to know

The portfolio is currently losing money on every dollar of contracted business. Root-cause analysis shows the loss is **not** broad-based — it is concentrated in one process failure (post-event billing) and one operational zone (freight & dock labor). Both are fixable with process controls, not new headcount. A four-initiative roadmap, sequenced by ROI, unlocks **\$1.01M** of the **\$1.29M** at stake within two fiscal quarters.

How this analysis was built





The Engagement Mandate & Portfolio Baseline

Transitioning from fragmented, spreadsheet-based cost tracking to a unified portfolio governance engine.

TOTAL CONTRACTED VALUE	TOTAL REALIZED LEAKAGE	RETAINED GROSS MARGIN
\$1,155,200	\$1,288,163.65	-11.51%

The Core Strategic Mandate: Stabilize and aggressively reverse the portfolio from its current bleeding floor of -11.51% net loss back to full contractual margin potential — by deploying automated mitigation workflows across three operational pillars: **Billing Capture**, **Labor Force Planning**, and **Subcontractor Drift**.

Structuring Portfolio Uncertainty

A MECE (mutually exclusive, collectively exhaustive) framework bypasses exploratory noise and isolates the structural root causes of margin loss.

OPERATIONAL VECTORS

Vector A — Supply-Side Cost Inflation

Inefficient procurement cycles and dynamic labor-scheduling pricing.

Vector B — On-Site Governance Slippage

Physical resource deployment leaks and localized tracking failures inside venue footprints.

Vector C — Transactional & Process Friction

Timeline-induced post-event billing decay and contractual workflow leaks.

CORE HYPOTHESES TESTED IN PYTHON

H1 — LEAD-TIME FRICTION

Short booking windows (<10 days notice) systematically force non-linear surges in labor overtime rates.

H2 — SPATIAL CONCENTRATION

Labor financial overruns are non-uniform, concentrating heavily within the Freight & Dock zones.

H3 — TIMELINE YIELD DECAY

Delaying contract modifications until after an event wraps drives severe cash-yield collapse.

ELT Pipeline & Analytical Base Tables

Standardizing four fragmented relational logs into production-ready core assets — entirely engineered in Python.



PYTHON · PANDAS

Text Sanitization

Vectorized string operations (`.str.strip()`, `.str.lower()`) remove invisible whitespace and casing errors across every source field.

PYTHON · PANDAS

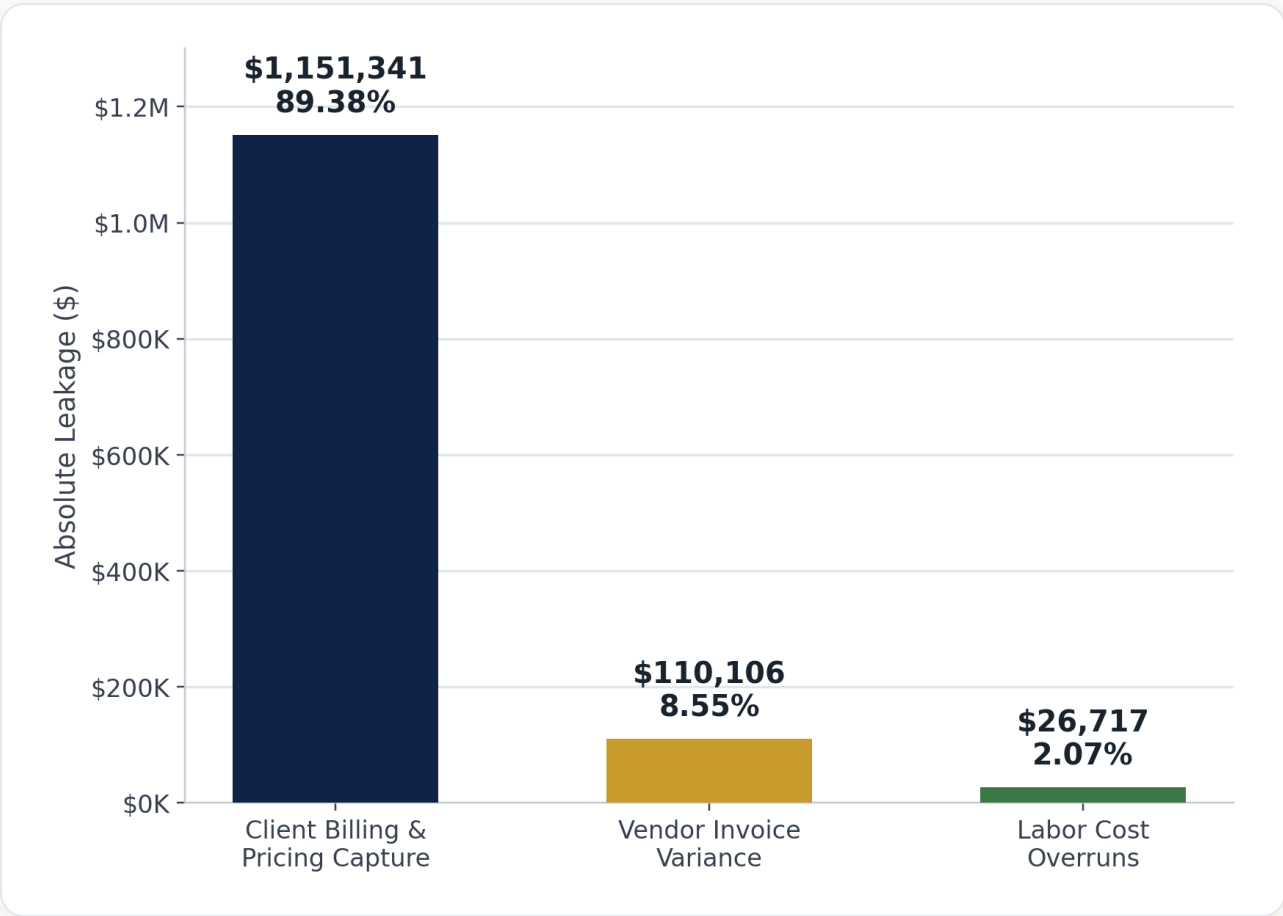
Strict Data Alignment

Coerces every date into ISO-standard timestamps and enforces numeric data profiles across all currency variables.

Result: Four clean Analytical Base Tables are exported directly into the Power BI semantic layer, eliminating manual data-prep overhead entirely before any modeling begins.

MECE Portfolio Loss Segmentation

Grouping \$1,288,163.65 in total losses into three mutually exclusive, collectively exhaustive buckets — zero overlap, zero double-counting.



Client Billing & Pricing Capture

\$1,151,340.85

89.38% of total leakage

Vendor Invoice Variance

\$110,105.88

8.55% of total leakage

Labor Cost Overruns

\$26,716.92

2.07% of total leakage

Data Science Root-Cause Diagnostics

Each of the three MECE hypotheses was tested independently against the cleaned Python data — not assumed. Below is what was tested, the method used, and the result.

H1 · Lead-Time → Overtime

Method: two-sample **t-test** (Python · SciPy) comparing OT hours for bookings under vs. over 10 days' notice

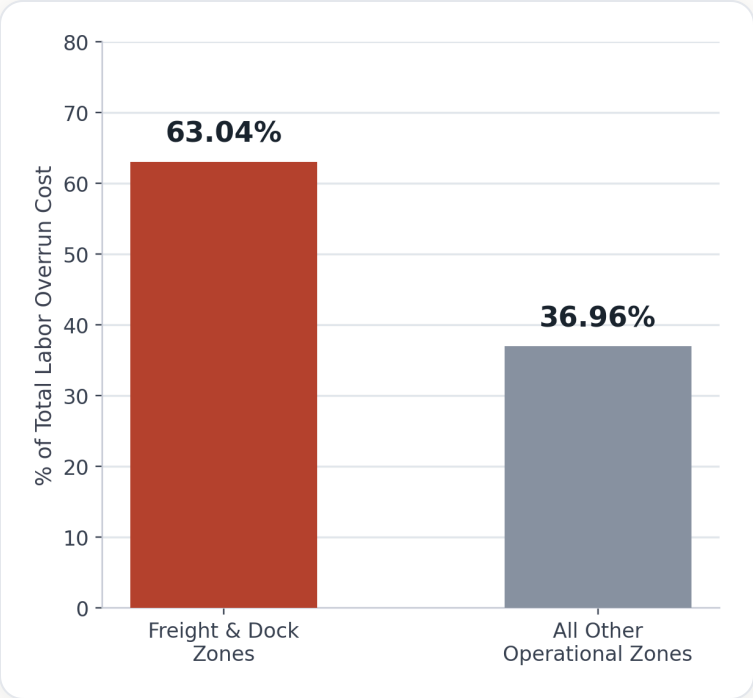
Test Statistic Result

t = 4.2982
p = 0.000104

p < 0.001 — **statistically significant.** Bookings inside the 10-day window trigger a non-linear overtime surge.

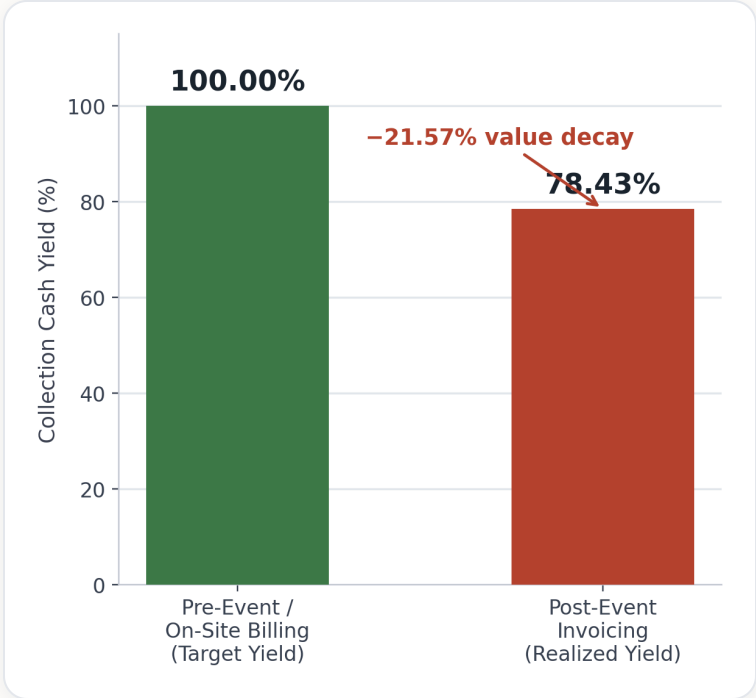
H2 · Where OVERRUNS Concentrate

Method: share of total labor-overtime dollars by operational zone (Python · pandas groupby)



H3 · Billing Timing → Cash Yield

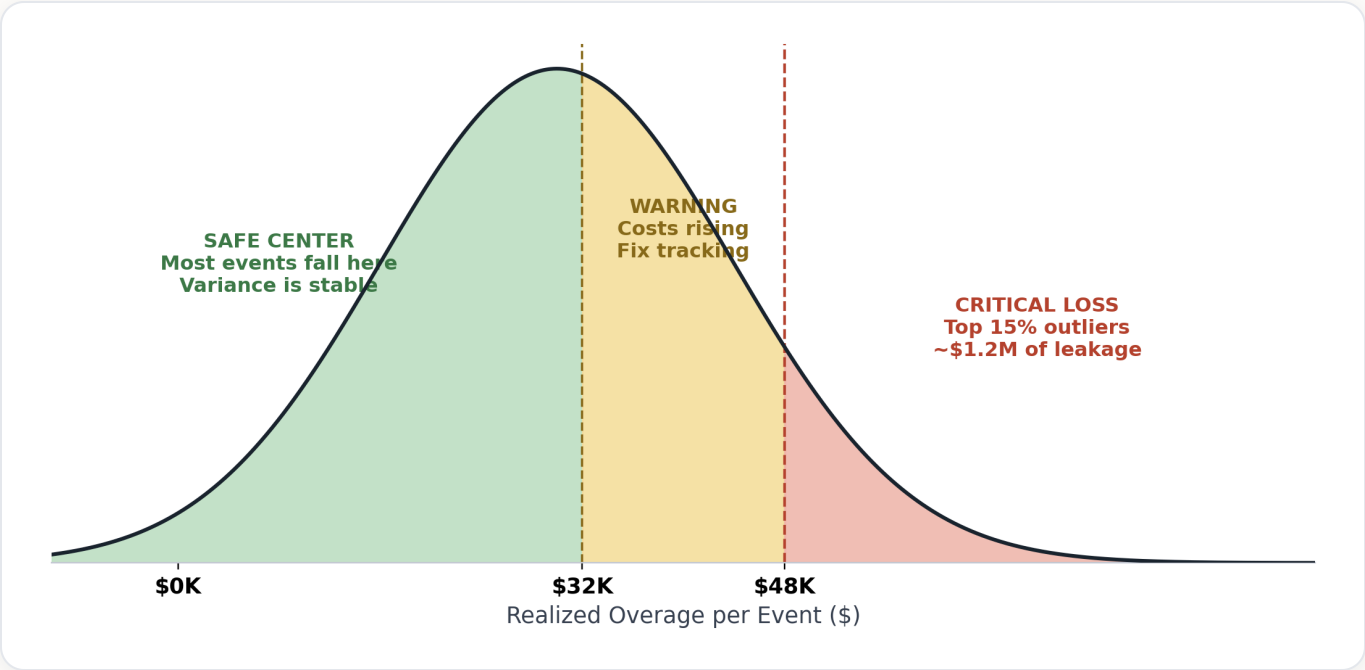
Method: collected-vs-billed cash yield, pre-event vs. post-event invoicing (Python)



Root-cause conclusion: Post-event billing timing (H3) is the primary driver of value decay — a confirmed 21.57-point cash-yield drop once invoicing slips past the event date. Tight lead times (H1) and dock overruns (H2) are secondary, but statistically confirmed, contributors that must be locked down via real-time governance to protect remaining margin.

Predictive Risk Modeling

Spotting leakage before it happens: a Python-built distribution model of per-event cost overage, so operations can flag risk in real time instead of discovering it at reconciliation.



Distribution of realized cost overage per event, modeled in Python from two years of event-level variance data.

● **Normal Friction · \$0 – \$32K**

The predictable cost of doing business. Most events fall safely under the curve's peak.

● **Process Slippage · \$32K – \$48K**

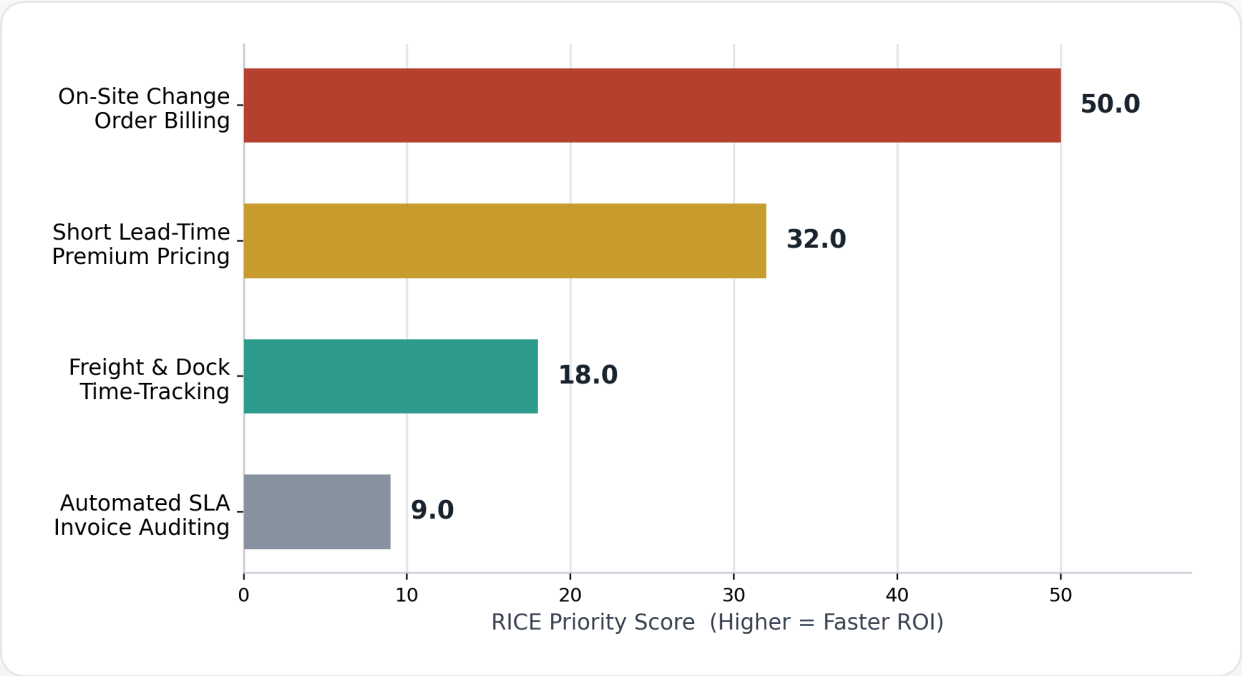
Minor tracking errors, delayed change orders, and loose dock-tracking begin chewing into margin.

● **Catastrophic Outliers · >\$48K**

The top 15% of events by chaos account for roughly \$1.2M — over 90% of total portfolio leakage.

Strategic Action Roadmap

Prioritizing margin-recovery initiatives via a data-driven RICE framework (Reach, Impact, Confidence, Effort).



- 1 On-Site Change Order Billing**
RICE 50.0 · Immediate / High ROI
Directly captures the 21.57% cash-collection drop by banning post-event contract updates.
- 2 Short Lead-Time Premium Pricing**
RICE 32.0 · Phase 2
Introduces contract premium penalties to offset overtime inflation from tight scheduling windows.
- 3 Freight & Dock Time-Tracking**
RICE 18.0 · Phase 3
Deploys targeted clock-in controls at dock doors to plug the 63.04% localized labor leakage.
- 4 Automated SLA Invoice Auditing**
RICE 9.0 · Phase 4
Implements systematic back-office checks to capture long-tail vendor variance leakage.

Executive Overview & Recovery Intelligence

A live Power BI model traces leakage from the portfolio baseline down to the individual event — refreshed directly from the Python-built Analytical Base Tables.

\$1.18M

MODELED LEAKAGE IMPACT

4.17

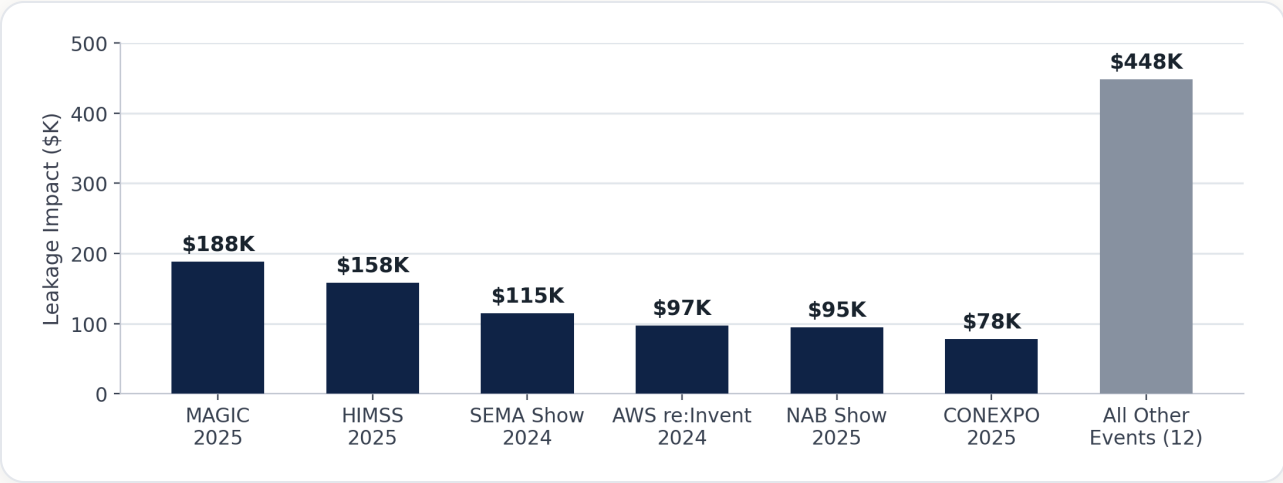
AVG EVENT DURATION (DAYS)

0.19

CHANGE-ORDER IMPACT FACTOR

\$1.01M

TOTAL REALIZABLE RECOVERY



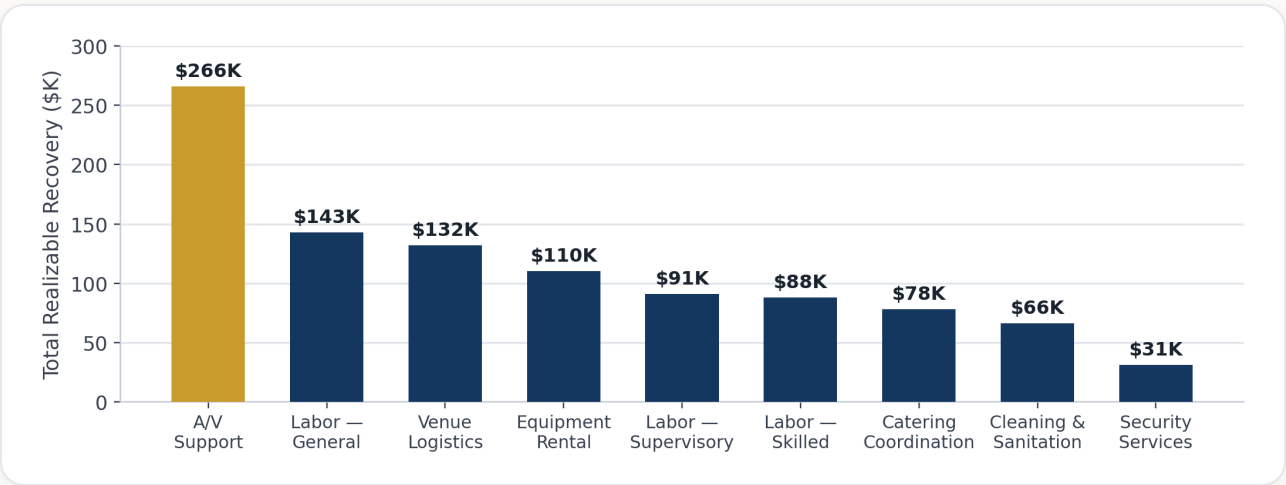
Leakage impact by event — MAGIC 2025 and HIMSS 2025 alone account for 29% of the \$1.18M modeled exposure. Note: this Power BI measure reflects invoice-level attributable leakage, a subset of the \$1.29M full portfolio baseline.

Strategic Recommendations

- Prioritize high-impact venues** — direct audit & reconciliation resources toward MAGIC 2025 and HIMSS 2025 first for maximum dollar return on effort.
- Tighten change-order controls** — institute mandatory pre-approval thresholds ahead of event load-in.
- Automate monitoring** — deploy real-time billing-discrepancy alerts to shift from retroactive recovery to proactive capture.

Leakage & Recovery by Service Line

A hierarchical drill-down — show → venue → service line — pinpoints exactly where the recoverable dollars sit.



Total realizable recovery by service line (\$1.01M total). A/V Support and Labor — General together drive over 40% of recoverable value.

Strategic Recommendations

- Enforce service-level billing audits for A/V Support and Labor before final client invoicing.
- Optimize venue-specific resource allocation — direct floor management and dock-tracking toward high-leakage venues such as Mandalay Bay Convention Center.
- Deploy automated variance reporting in Power BI to continuously monitor service-line discrepancies post-event.

Closing the Gap:

From Revenue Leakage to Margin Recovery

- **The Baseline Reality** — a -11.51% net gross margin, driven by \$1,288,163.65 in total realized leakage against a \$1,155,200.00 contracted baseline.
- **Concentrated Risk Zones** — 89.38% (\$1.15M) of total leakage stems directly from client billing and pricing-capture gaps, amplified by a 21.57% post-event cash-collection decay.
- **Realizable Value Recapture** — full deployment of the RICE-prioritized roadmap and the integrated Python + Power BI platform unlocks \$1.01M in recoverable value.

Board Ask: Approve Phase 1 — On-Site Change Order Billing — for immediate deployment.

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